

Dynamic Analysis of the Poisson Paradox: Path B Structural Evolution

DOCUMENT VI — EMPIRICAL VALIDATION, SAFETY METRICS & RISK CONTROLS

1. Operational Performance & Empirical Validation

Longitudinal empirical testing of the Reverse Engineering branch over the historical Serie A dataset validates the structural persistency of the overshoot anomaly. The model filters out team-specific variance by isolating purely algorithmic inefficiencies within institutional software, achieving consistent performance across multiple seasons.

Match Event	Pre-Match CLV	Goal Minute ()	Model Fair Odds	Market Peak Spike	Extracted Edge (+EV)
Genoa vs. Lazio	1.62	11'	~2.55	3.50 – 3.55	+34.3% Alpha Extracted
Lecce vs. Cagliari	1.53	8'	~2.15	2.85 – 2.95	+30.0% Alpha Extracted
Verona vs. Cagliari	1.53	14'	~2.09	2.65 – 2.80	+29.1% Alpha Extracted
Torino vs. Cremonese	1.67	26' (Safety Stop)	2.10	2.25 – 2.30	<5% (Position Rejected)

The empirical results demonstrate a highly structured return profile, with captured alpha fluctuating between 29.1% and 34.3% inside the optimal Strike Zone. Positions initiated outside these boundaries are automatically neutralized by the system's temporal constraints.

2. Risk Management & Portfolio Stabilization

To eliminate capital exposure during overdispersion clusters (periods of high goal frequency), the framework utilizes a tiered risk architecture designed to prevent downside capitalization degradation:

- **Fractional Kelly Scaling:** Active entries inside the high-volatility live paths (Paths A and B) are strictly capped at $\frac{1}{4}$ *Kelly*. Terminal positions focused on late-game time decay (Path C) operate at $\frac{1}{8}$ *Kelly*.
- **Monte Carlo & Block Bootstrapping:** Drawdown boundaries are calculated through 10,000-iteration Monte Carlo simulations using block bootstrapping methods to preserve underlying temporal dependencies. These simulations define a strict Maximum Drawdown (MDD) ceiling of 17.89%.

3. The Constraint Flip Automation

The core layer of capital protection is governed by the "Constraint Flip" logic gate. The portfolio tracking engine continuously monitors current drawdown trajectories against the Monte Carlo variance bounds:

If Drawdown approaches the 17.89% threshold:
Trigger Gate $\rightarrow$ *Freeze Path B Live Operations* $\rightarrow$ *Re-route 100% Capital to Path C Rent Extraction*

By shifting portfolio capital away from speculative live shocks and reallocating it entirely to the low-volatility, 91% success rate terminal decay markets of Path C, the system guarantees principal insulation and automates bankroll recovery without requiring human emotional intervention.